

Refinements Open Ended Assignment

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Refinement 1 - Begin Graphing in RStudio Open Ended Assignment 1

Question 1: Does a person having a child under 18 influence how often they fly annually?

```
require(ggmosaic)
```

```
## Loading required package: ggmosaic
```

```
## Warning: package 'ggmosaic' was built under R version 4.4.3
```

```
data(fly)
```

```
head(fly)
```

```
## # A tibble: 6 x 27
```

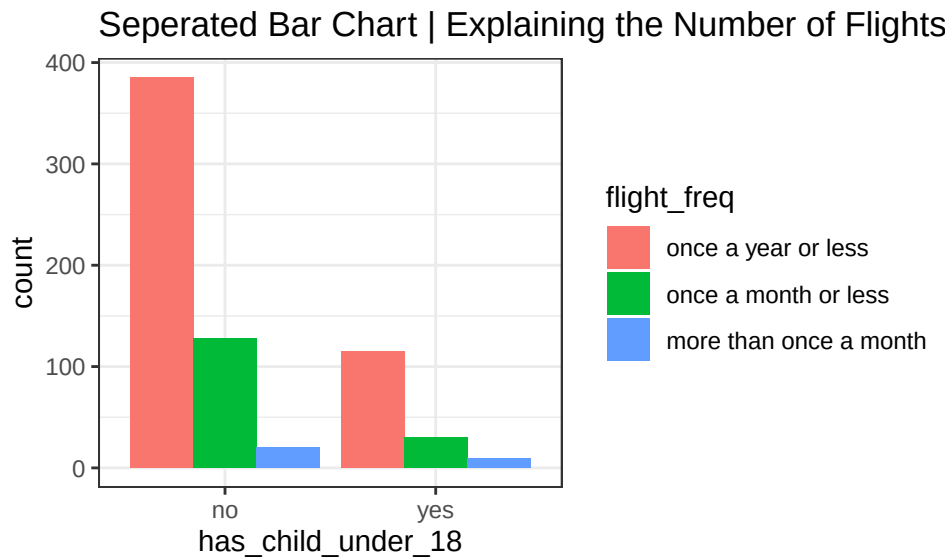
```
##       id flight_freq      do_you_recline      height has_child_under_18
##       <dbl> <fct>          <fct>          <fct>    <fct>
## 1 3436139758 once a year or less <NA>          <NA>    <NA>
## 2 3434278696 once a year or less about half the time "6'3\" yes
## 3 3434275578 once a year or less usually          "5'8\" no
## 4 3434268208 once a year or less always          "5'11\"~ no
## 5 3434250245 once a month or less about half the time "5'7\" no
## 6 3434245875 once a year or less usually          "5'9\" yes
## # i 22 more variables: three_seats_two_arms <fct>, two_seats_one_arm <fct>,
## #   window_shade <fct>, rude_to_move_to_unsold_seat <fct>,
## #   rude_to_talk_to_neighbor <fct>, six_hr_flight_leave_seat <fct>,
## #   reclining_obligation_to_behind <fct>, rude_to_recline <fct>,
## #   eliminate_reclining <fct>, rude_to_switch_seats_friends <fct>,
## #   rude_to_switch_seats_family <fct>, rude_to_wake_neighbor_bathroom <fct>,
## #   rude_to_wake_neighbor_walk <fct>, rude_to_bring_baby <fct>, ...
```

```
fly=na.omit(fly)
```

```
print("The Following is my Original Plot from Begin Graphing in RStudio Open Ended")
```

```
## [1] "The Following is my Original Plot from Begin Graphing in RStudio Open Ended"
```

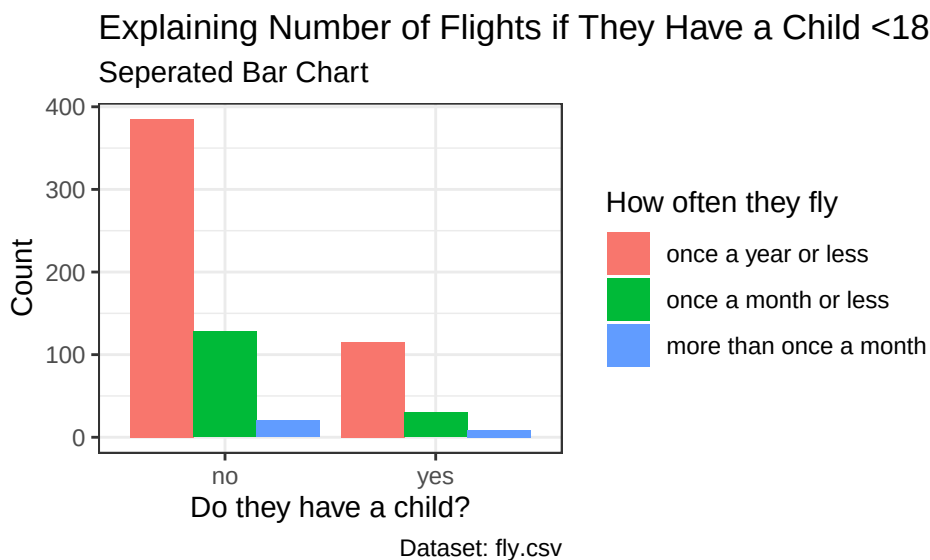
```
gf_bar( ~ has_child_under_18 , data = fly , fill = ~ flight_freq, position = position_dodge( ), title="S
```



```
print("The Following is my Revised Plot from Begin Graphing in RStudio Open Ended")
```

```
## [1] "The Following is my Revised Plot from Begin Graphing in RStudio Open Ended"
```

```
gf_bar( ~ has_child_under_18 , data = fly , fill = ~ flight_freq, position = position_dodge( ))>%gf_lab
```



I feel like my intial graph was very strong but I felt it was necessary to revise it because it had flight_freq which made it gross and the axes labels were very unappealing. Now I feel like it is significantly better but I still wonder if there is a better title that could be used. Its hard to incorporate it without it being too long. Perhaps if there is a way to wrap text but even then that may be too wordy and take away from the graph.

Refinement 2 - Begin Graphing in RStudio Open Ended Assignment 2

Question 2: Is there a relationship between a State's Per Capita Income and the Winner of the 2008 Election?

```
require(Stat2Data)
```

```
## Loading required package: Stat2Data
```

```
data(Election08)
```

```
head(Election08)
```

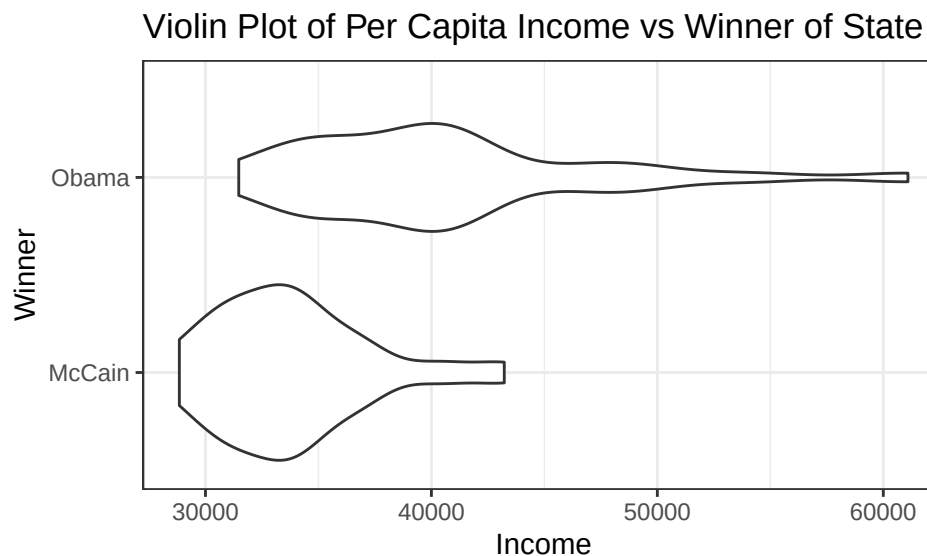
```
##      State Abr Income  HS  BA Dem.Rep ObamaWin
## 1  Alabama  AL  32404 80.4 21.4    -1        0
## 2   Alaska  AK  40352 90.5 26.0   -11        0
## 3   Arizona  AZ  33029 83.5 25.3     0        0
## 4  Arkansas  AR  30060 81.1 19.3    12        0
## 5 California CA  41571 80.2 29.5    19        1
## 6  Colorado  CO  41042 88.9 35.0    11        1
```

```
Election08$Winner <- ifelse(Election08$ObamaWin == 1, "Obama", "McCain")
```

```
print("The Following is my Original Plot from Begin Graphing in RStudio Open Ended")
```

```
## [1] "The Following is my Original Plot from Begin Graphing in RStudio Open Ended"
```

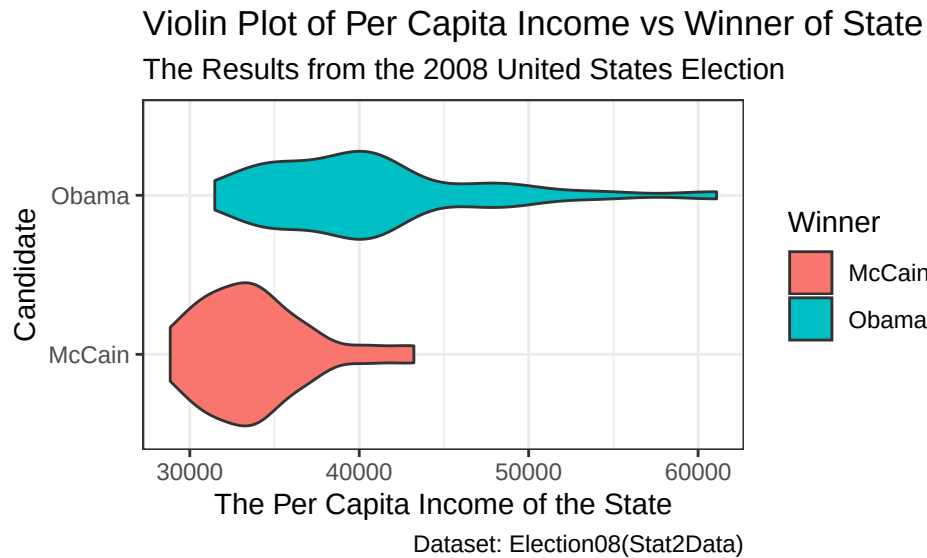
```
gf_violin( Winner ~ Income , data = Election08 ,title="Violin Plot of Per Capita Income vs Winner of State")
```



```
print("The Following is my Revised Plot from Begin Graphing in RStudio Open Ended")
```

```
## [1] "The Following is my Revised Plot from Begin Graphing in RStudio Open Ended"
```

```
gf_violin( Winner ~ Income , data = Election08, fill = ~Winner)%>%gf_labs(x = "The Per Capita Income of States")
```

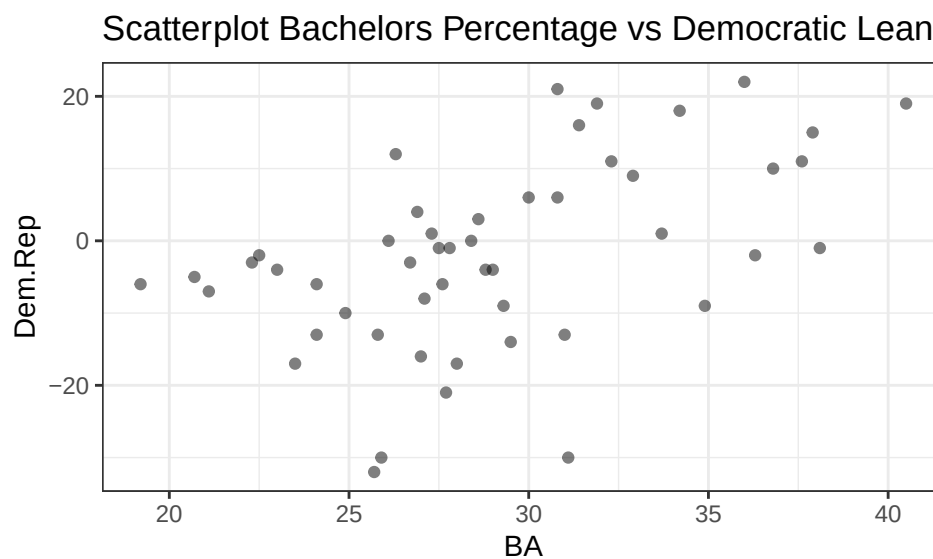


I feel like the difference between the two is night and day fully. Not only are the individual violins filled in with the proper party colors but this whole graph looks relatively professional. It shows the candidates along side the per capita income and clearly labeled key and title. I am very content with this one.

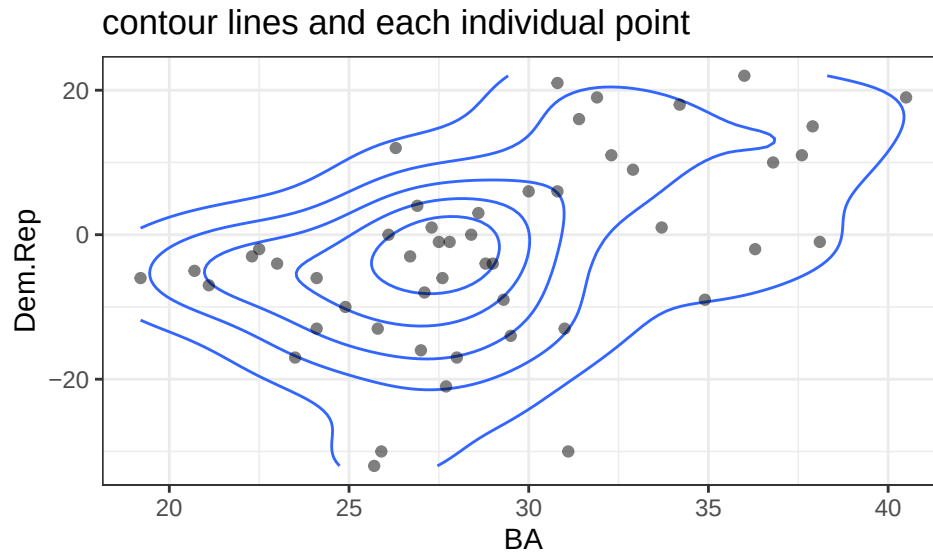
The New Graphs

Is there a relationship between the percent of College Graduates and Democratic Lean in the 2016 Election?

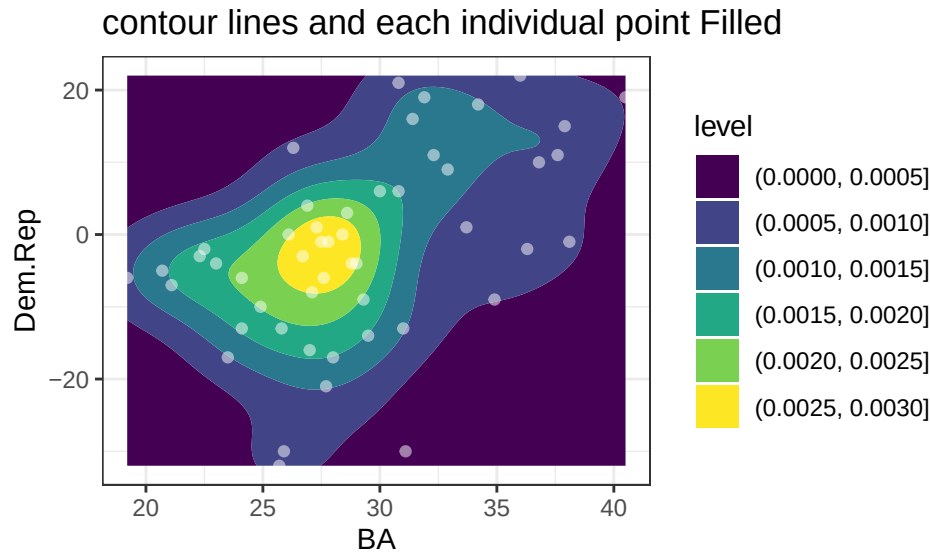
```
#numericvnum
require(Stat2Data)
data(Election16)
gf_point( Dem.Rep ~ BA , data = Election16 , alpha = 0.5 , title="Scatterplot Bachelors Percentage vs D
```



```
gf_density_2d( Dem.Rep ~ BA , data = Election16, title="contour lines and each individual point") %>% g
```



```
gf_density_2d_filled( Dem.Rep ~ BA , data = Election16, title="contour lines and each individual point Filled") %>% g
```



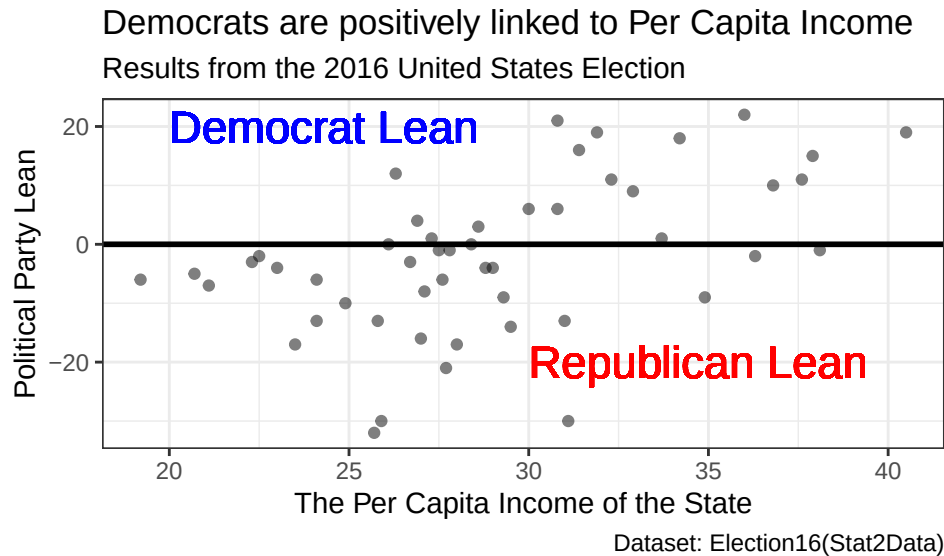
#REVISED

```
gf_point( Dem.Rep ~ BA , data = Election16 , alpha = 0.5) %>% gf_labs(x = "The Per Capita Income of the S
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: All aesthetics have length 1, but the data has 50 rows.
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
## All aesthetics have length 1, but the data has 50 rows.
```

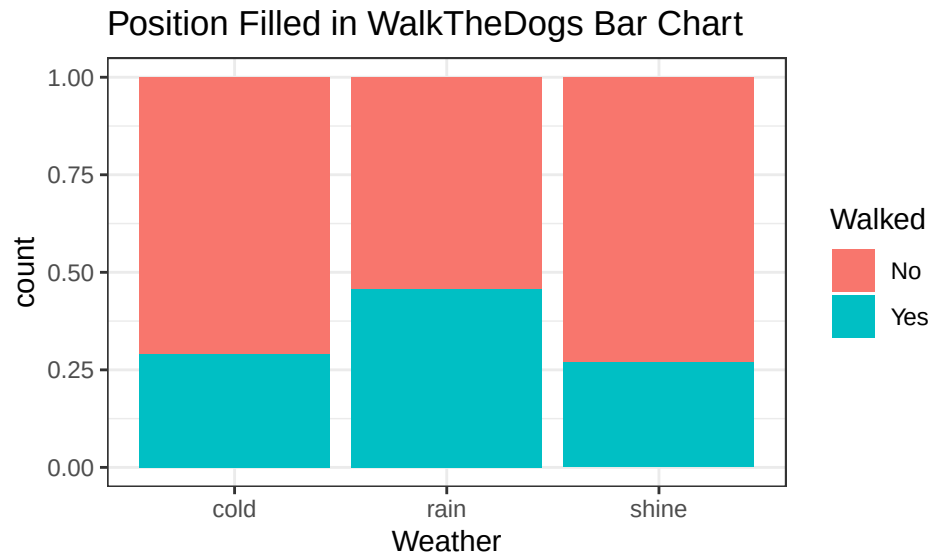
```
## i Please consider using `annotate()` or provide this layer with data containing
## a single row.
```



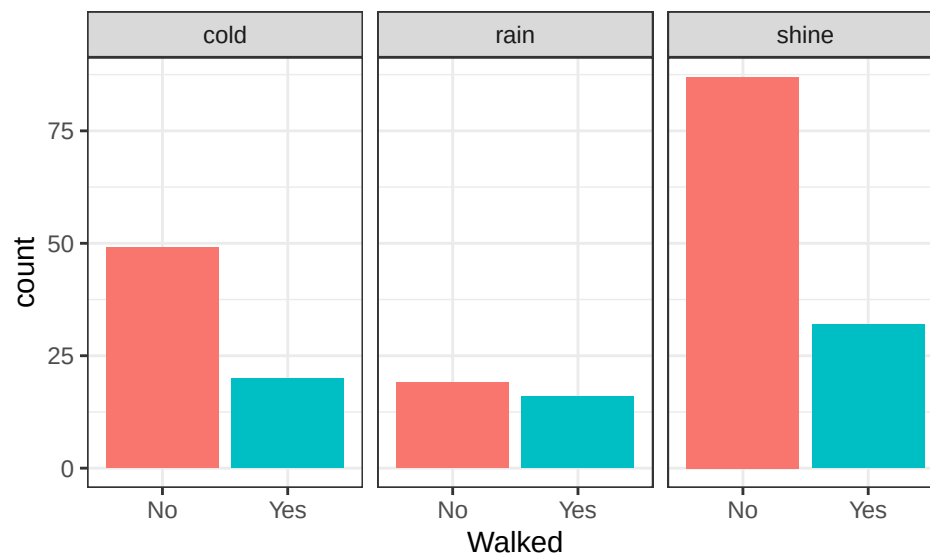
On this new graph, I started very bare bones but landed on this one as I felt like it illustrated an interesting design comparable to a win delta where its expected to be 0 for no relation. However, with this graph there was a clear relation so I did some touch ups. I added a horizontal line in to show the no relation line and added labels to show case what being above and below would mean. I also added labels and titles to make it accessible to the reader and giving a caption with the dataset and where it was accessed from.

Is there a relationship between the weather and if the dogs were walked?

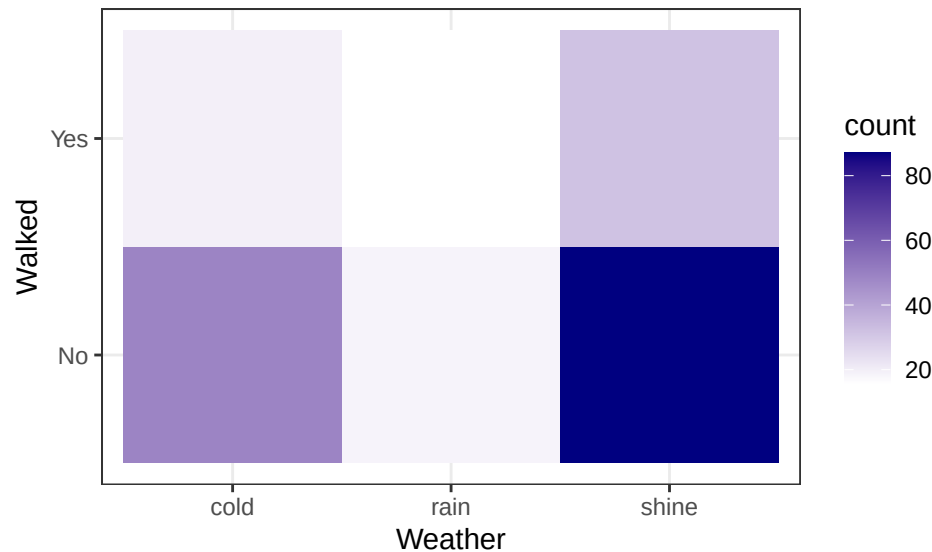
```
#CATEGORICAL
data(WalkTheDogs )
WalkTheDogs$Walked <- ifelse(WalkTheDogs$Walk == 1, "Yes", "No")
gf_bar( ~ Weather , data = WalkTheDogs , fill = ~ Walked , position = position_fill( ),title="Position")
```



```
gf_bar(~Walked, data = WalkTheDogs , fill = ~Walked, show.legend = FALSE) %>% gf_facet_wrap(~Weather)
```



```
gf_bin2d(Walked~Weather, data=WalkTheDogs ) %>%
gf_refine(scale_fill_continuous(low = "white", high = "navy"))
```

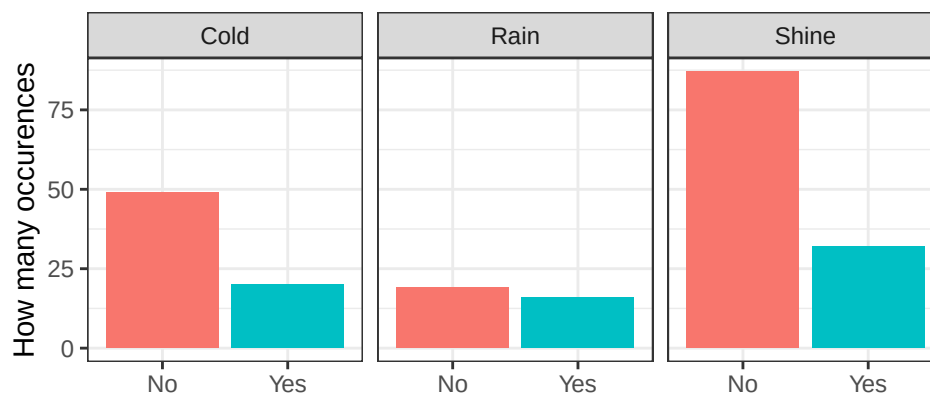


#REVISED

```
gf_bar(~Walked, data = WalkTheDogs, fill = ~Walked, show.legend = FALSE) %>% gf_facet_wrap(~Weather, 1)
```

Not Walking the Dog is Common in Any Weather

Bar Graph Seperated by Weather



The Weather and If the Dog was Walked

Dataset: WalkTheDogs(Stat2Data)

I decided to go with this graph because I felt like it provided the most interested and compelling story. The first position filled one was certainly interesting but I felt like it told an incomplete story. I added labels with a good title to explain the story to those who were curious about how often the dog is walked.

Is there a relationship between the rater's sex and how they rate someone's sexual dimorphism?

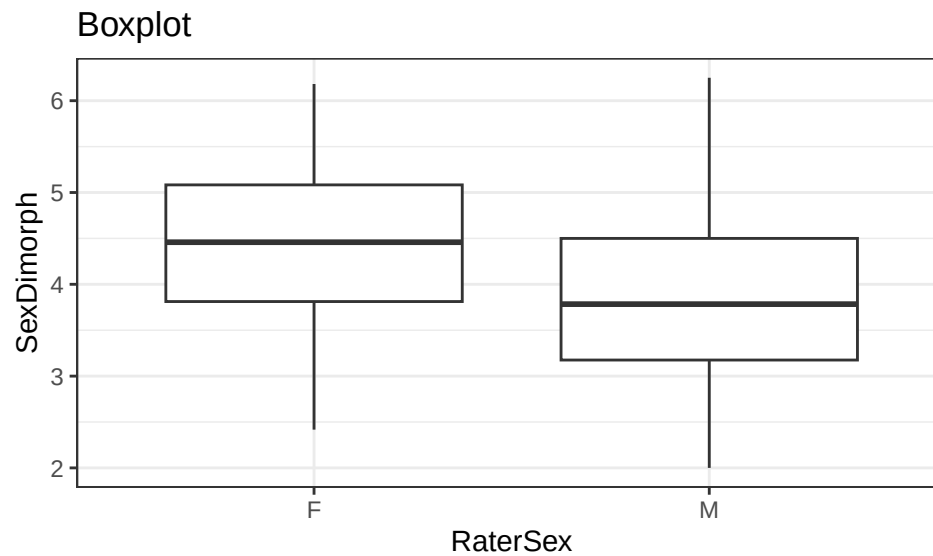
#1n1

```
data(FaithfulFaces)
```

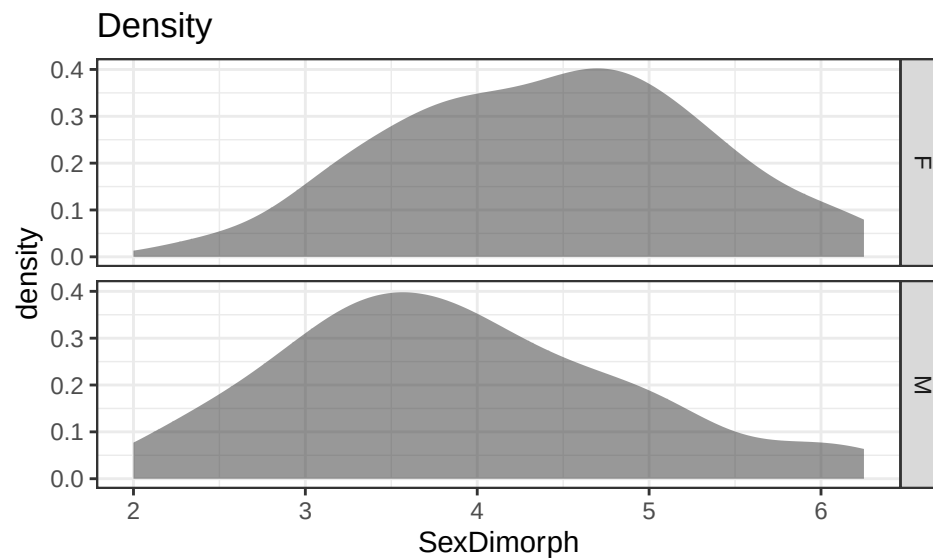
```
library(ggribes)
```

```
## Warning: package 'ggribes' was built under R version 4.4.3
```

```
gf_boxplot( SexDimorph ~ RaterSex , data = FaithfulFaces , options,title="Boxplot")
```

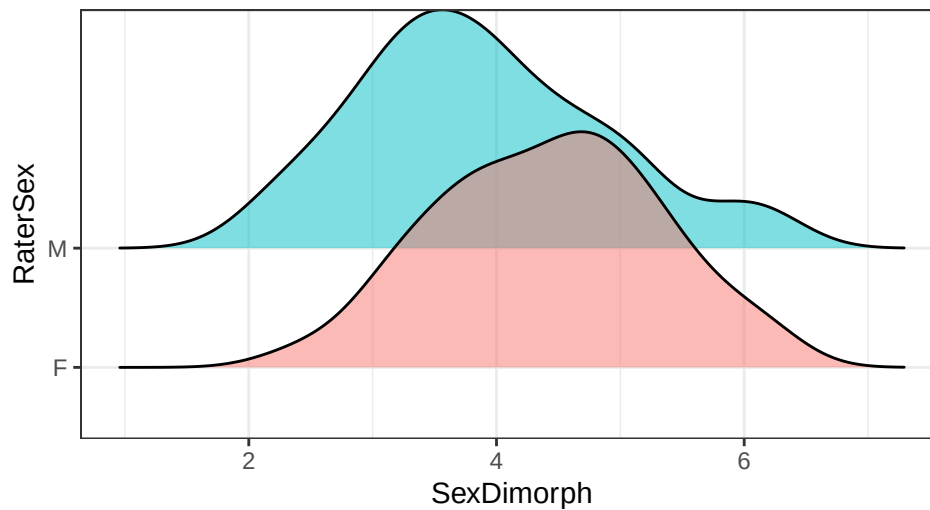


```
gf_density(~SexDimorph, data = FaithfulFaces,title="Density") %>%
gf_facet_wrap(~RaterSex, ncol=1, strip.position = "right")
```



```
gf_density_ridges(RaterSex~SexDimorph, data=FaithfulFaces, fill=~RaterSex, show.legend = FALSE, scale =
## Picking joint bandwidth of 0.347
```

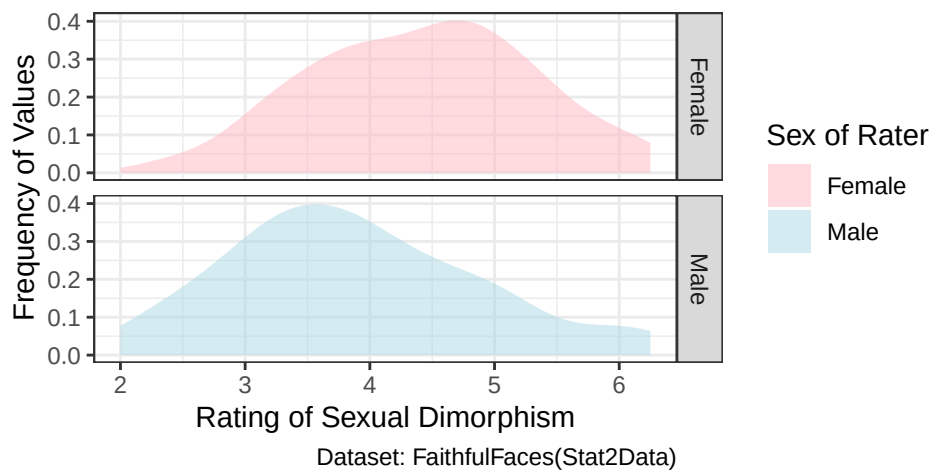
Density Ridges Plot



```
#
#REVISED
gf_density(~SexDimorph, data = FaithfulFaces, fill=~RaterSex) %>% gf_facet_wrap(~RaterSex, ncol=1, strip=TRUE)
```

Women Rate Higher Sexual Dimorphism than Men

Density of Sexual Dimorphism Separated by Sex



This density plot was the most intriguing when I did this as it gave me a unique way to look at how sexual dimorphism is perceived and the proportion at each part. This graph was very fun to edit and add on to. Obviously it was labelled on the axes and titles as with the others. I also labeled the key and gave colors to each density plot making it look nice. I was very happy with the way it turned out and it was worth the struggles with the code.